

Linear Models Lecture 11: Asymptotic theory for OLS

Robert Gulotty

University of Chicago

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Today's Claims

- The least squares estimator $\hat{\beta}$ is *consistent* for the projection coefficient β .
- There is an asymptotic normal approximation to the distribution of $\hat{\beta}$ (once normalized).
- Similarly, the error variance estimators $\hat{\sigma}^2$ and s^2 are consistent for σ^2 .
- There is a consistent estimator of $\mathbf{V}_\beta$, either under homoskedasticity $\hat{\mathbf{V}}_\beta^0$, or otherwise.
- We can use z-statistics for asymptotic standard errors, confidence intervals, and prediction intervals.

Strategy: Apply the tools from Lecture 10 — WLLN, CLT, CMT — plus the delta method (introduced today) to derive the asymptotic properties of OLS.

Motivation: Distribution of a Function of Estimators

- Suppose we want to estimate some function of parameters $f(b_1, b_2) = b_1/b_2$.
- What is the distribution of $\mathbf{f}(\mathbf{b})$?
- We can use the **Delta Method** (Asymptotic Normality + Taylor Expansion) to approximate it.
- The idea: if $\mathbf{b}$ is approximately normal (by the CLT), then a smooth function $\mathbf{f}(\mathbf{b})$ is also approximately normal, with variance determined by the derivatives of f .

Asymptotic Normality of $\mathbf{b}$ (Preview)

Claim (proved later this lecture): if the errors are independently distributed with mean zero and finite variance and $\mathbf{X}$ is of full rank, the CLT gives us:

$$\mathbf{b} \overset{a}{\sim} N[\boldsymbol{\beta}, \frac{\sigma^2}{n} [\text{plim} \frac{\mathbf{X}'\mathbf{X}}{n}]^{-1}]$$

That is:

$$\text{Asy. Var}(\mathbf{b} - \boldsymbol{\beta}) = \sigma^2 \frac{[\text{plim} \frac{\mathbf{X}'\mathbf{X}}{n}]^{-1}}{n}$$

Which we will estimate with:

$$s^2(\mathbf{X}'\mathbf{X})^{-1} = \frac{\mathbf{e}'\mathbf{e}}{n - K}(\mathbf{X}'\mathbf{X})^{-1}$$

For now, take this as given. Once $\mathbf{b}$ is approximately normal, we can ask: what is the distribution of a *function* of $\mathbf{b}$?

The Delta Method

- Define a $J \times K$ matrix of partial derivatives across J functions and K variables.

$$\begin{aligned}\mathbf{C}(\mathbf{b}) &= \frac{\partial \mathbf{f}(\mathbf{b})}{\partial \mathbf{b}'} \\ \text{plim} \mathbf{C}(\mathbf{b}) &= \text{plim} \frac{\partial \mathbf{f}(\mathbf{b})}{\partial \mathbf{b}'} \\ &= \frac{\partial \mathbf{f}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} \equiv \boldsymbol{\Gamma} \quad (\text{Slutsky})\end{aligned}$$

- The Taylor approximation from $\boldsymbol{\beta}$ to $\mathbf{b}$ is then

$$\mathbf{f}(\mathbf{b}) = \mathbf{f}(\boldsymbol{\beta}) + \boldsymbol{\Gamma} \times (\mathbf{b} - \boldsymbol{\beta}) + O_p(n^{-1})$$

- The asymptotic distribution is: $\mathbf{f}(\mathbf{b}) \overset{a}{\sim} N(\mathbf{f}(\boldsymbol{\beta}), \boldsymbol{\Gamma}[\text{Asy. Var}(\mathbf{b} - \boldsymbol{\beta})]\boldsymbol{\Gamma}')$
- We can estimate this approximation with: $\mathbf{C}[s^2(\mathbf{X}'\mathbf{X})^{-1}]\mathbf{C}'$

Application: Greene Example 4.4

- Suppose we think effects persist through time:

$$y_{it} = \beta_0 + \beta_1 x_{it} + \gamma y_{it-1} + \epsilon_{it}$$

- In the long run we expect stability:

$$\bar{y} = \beta_0 + \beta_1 \bar{x} + \gamma \bar{y}$$

$$\bar{y} - \gamma \bar{y} = \beta_0 + \beta_1 \bar{x}$$

$$\bar{y}(1 - \gamma) = \beta_0 + \beta_1 \bar{x}$$

$$\bar{y} = \frac{\beta_0}{(1 - \gamma)} + \frac{\beta_1}{(1 - \gamma)} \bar{x}$$

Long Run Effects with Lagged DV

- Suppose we raise x_{it} by 1:

$$\begin{aligned}y_{it}^* &= \beta_0 + \beta_1(\bar{x} + 1) + \gamma\bar{y} &&= \bar{y} + \beta_1 \\y_{it+1}^* &= \beta_0 + \beta_1\bar{x} + \gamma(\bar{y} + \beta_1) &&= \bar{y} + \gamma\beta_1 \\y_{it+2}^* &= \beta_0 + \beta_1\bar{x} + \gamma(\bar{y} + \gamma\beta_1) &&= \bar{y} + \gamma^2\beta_1\end{aligned}$$

- The long run effect of a temporary one unit increase in x is thereby:

$$\begin{aligned}&= \beta_1 + \gamma\beta_1 + \gamma^2\beta_1 + \gamma^3\beta_1 + \dots \\&= \beta_1(1 + \gamma + \gamma^2 + \gamma^3 + \dots) = \beta_1 \frac{1}{1 - \gamma}\end{aligned}$$

Example: Gas Consumption

- Demand for gasoline is based on the kind of vehicle consumers choose to buy and how much they drive.
- We can model consumer demand for gas (G =gas expenditures/gas price) as a function of gas prices, income (Y) and past car purchasing decisions, which are themselves a function of gas prices and income.

$$\ln(G/pop)_t = \hat{\beta}_1 + \hat{\beta}_2 \ln P_{G,t} + \hat{\beta}_3 \ln(Y/pop)_t + \hat{\beta}_4 \ln(G/Pop)_{t-1} + e_t$$

- $\hat{\beta}_2$ is the short run price elasticity. $\hat{\beta}_3$ is the short run income elasticity.
- $\frac{\hat{\beta}_2}{1-\hat{\beta}_4}$ is the long run price elasticity, $\frac{\hat{\beta}_3}{1-\hat{\beta}_4}$ is the long run income elasticity.

Delta Method Applied to Gas Consumption

■ Call $\mathbf{b} = (\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3, \hat{\beta}_4)$, $\mathbf{f}(\mathbf{b}) = \begin{bmatrix} \frac{\hat{\beta}_2}{1-\hat{\beta}_4} \\ \frac{\hat{\beta}_3}{1-\hat{\beta}_4} \end{bmatrix}$

$$\mathbf{C}(\mathbf{b}) = \frac{\partial \mathbf{f}(\mathbf{b})}{\partial \mathbf{b}'} = \begin{bmatrix} \frac{\partial \frac{\hat{\beta}_2}{1-\hat{\beta}_4}}{\partial \hat{\beta}_1} & \frac{\partial \frac{\hat{\beta}_2}{1-\hat{\beta}_4}}{\partial \hat{\beta}_2} & \frac{\partial \frac{\hat{\beta}_2}{1-\hat{\beta}_4}}{\partial \hat{\beta}_3} & \frac{\partial \frac{\hat{\beta}_2}{1-\hat{\beta}_4}}{\partial \hat{\beta}_4} \\ \frac{\partial \frac{\hat{\beta}_3}{1-\hat{\beta}_4}}{\partial \hat{\beta}_1} & \frac{\partial \frac{\hat{\beta}_3}{1-\hat{\beta}_4}}{\partial \hat{\beta}_2} & \frac{\partial \frac{\hat{\beta}_3}{1-\hat{\beta}_4}}{\partial \hat{\beta}_3} & \frac{\partial \frac{\hat{\beta}_3}{1-\hat{\beta}_4}}{\partial \hat{\beta}_4} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{1-\hat{\beta}_4} & 0 & \frac{\hat{\beta}_2}{(1-\hat{\beta}_4)^2} \\ 0 & 0 & \frac{1}{1-\hat{\beta}_4} & \frac{\hat{\beta}_3}{(1-\hat{\beta}_4)^2} \end{bmatrix}$$

- The rows of this matrix are called gradient vectors.

```
USGasG <- read_csv("http://people.stern.nyu.edu/wgreene/Text/
                    Edition7/TableF2-2.csv")
USGasG <- USGasG %>%
  mutate( logG      = log(GASEXP/(GASP*POP)),
          logGlag   = lag(logG),
          logGASP   = log(GASP))
fm <- lm(logG ~ logGASP+log(INCOME)+log(PNC)+log(PUC)+logGlag,
        data= USGasG )
xprimex <- vcov(fm)
sigmasq <- sigma(fm)^2
bs <- coef(fm)
ggas <- c(0, 1/(1-bs[6]), 0, 0, 0, bs[2]/(1-bs[6])^2)
gincome <- c(0, 0, 1/(1-bs[6]), 0, 0, bs[3]/(1-bs[6])^2)
seGpricelongrun <- sqrt(t(ggas)%*(xprimex)%*ggas)
seIncomelongrun <- sqrt(t(gincome)%*(xprimex)%*gincome)
```

```
> cat("Long run elasticity to Gas prices (95% conf. int.):",  
      round(bs[2]/(1-bs[6]), 2), "+/-",  
      round(seGpricelongrun*qnorm(.975),2))  
Long run elasticity to Gas prices (95% conf. int.): -0.41 +/- 0.3  
  
> cat("Long run elasticity to Income (95% conf. intervals):",  
      round(bs[3]/(1-bs[6]), 2), "+/-",  
      round(seIncomelongrun*qnorm(.975),2))  
Long run elasticity to Income (95% conf. int.): 0.97 +/- 0.32
```

Canned Package Methods

In R:

```
> car :: deltaMethod(fm, "logGASP/(1-logGlag)")
```

	Estimate	SE	2.5 %	97.5 %
logGASP/(1 - logGlag)	-0.41136	0.15230	-0.70985	-0.1129

In STATA:

```
nlcom (_b[x2]/(1-_b[x6]))
```

Delta Method for Probit: Average Marginal Effects

Recall from Lecture 9: the Probit average marginal effect is a **nonlinear function** of $\hat{\beta}$:

$$\widehat{AME}_j = \frac{1}{n} \sum_{i=1}^n \phi(X_i' \hat{\beta}) \hat{\beta}_j$$

This is exactly the kind of problem the delta method solves. Define $h(\beta) = AME_j(\beta)$. Then:

$$\text{Asy. Var}(\widehat{AME}_j) = \nabla h(\hat{\beta})' \cdot \widehat{\text{Var}}(\hat{\beta}) \cdot \nabla h(\hat{\beta})$$

where ∇h accounts for both the direct effect ($\hat{\beta}_j$) and the indirect effect through $\phi(X_i' \hat{\beta})$.

Common pattern: Estimate $\hat{\theta}$ by MLE or OLS $\rightarrow$ compute a nonlinear function $h(\hat{\theta}) \rightarrow$ use the delta method for its standard error. Same logic for long-run elasticities, marginal effects, or any transformation of coefficients.

Target of Estimation

- We are after the projection coefficient $\beta = (\mathbb{E}[\mathbf{x}\mathbf{x}'])^{-1}\mathbb{E}[\mathbf{x}Y]$ from the projection model $Y = \mathbf{x}'\beta + e$
- We assume $(Y_i, \mathbf{x}_i)$ are i.i.d.
- And define $\mathbf{Q}_{XX} = \mathbb{E}[\mathbf{x}\mathbf{x}']$, assume is positive definite.
- We also need moments to exist, including 4th moments for asymptotic normality.

Proof Outline: Two Routes to Consistency

- **Route 1** (Method of Moments):
 - WLLN: sample moments converge to population moments
 - CMT: continuous functions of convergent sequences converge
 - OLS is a continuous function of sample moments
- **Route 2** (Algebraic):
 - Write $\hat{\beta} - \beta$ as a product of sample moments
 - Apply WLLN to each piece, CMT/Slutsky to combine

Both routes use the same tools from Lecture 10. Route 1 generalizes to nonlinear estimators (MLE, GMM); Route 2 gives an explicit formula for the sampling error.

OLS is a function of sample moments

- $\hat{\beta} = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i'\right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i Y_i\right) = \hat{\mathbf{Q}}_{XX}^{-1} \hat{\mathbf{Q}}_{XY}$
- $\hat{\mathbf{Q}}_{XX}$, $\hat{\mathbf{Q}}_{XY}$ are sample moments

Apply WLLN

- If $(\mathbf{x}_i, Y_i)$ are iid, then $\mathbf{x}_i \mathbf{x}_i'$, $\mathbf{x}_i Y_i'$ are iid.
- Recall from Lecture 10: WLLN says that if θ_i are iid with finite variance,
$$\frac{1}{n} \sum_{i=1}^n \theta_i \xrightarrow{P} \mathbb{E}[\theta]$$
- $\hat{\mathbf{Q}}_{XX} \xrightarrow{P} \mathbf{Q}_{XX}, \quad \hat{\mathbf{Q}}_{XY} \xrightarrow{P} \mathbf{Q}_{XY}$

Continuous Mapping Theorem (CMT)

- Recall from Lecture 10: if $Z_n \rightarrow_p c$ and $g(u)$ is continuous at c , then $g(Z_n) \rightarrow_p g(c)$.
- CMT tells us that plims survive continuous functions.
- $\hat{\beta} = g(\hat{Q}_{XX}, \hat{Q}_{XY})$ is a continuous function (matrix inversion is continuous at nonsingular matrices).
- $\hat{\beta} = \hat{Q}_{XX}^{-1} \hat{Q}_{XY} \rightarrow_p Q_{XX}^{-1} Q_{XY} = \beta$

Proof of Consistency of OLS: Route 2, Step 1

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \quad (\text{Definition})$$

$$= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \mathbf{e}) \quad (\text{Assumption 1})$$

$$= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e} \quad (\text{Inverse prop.})$$

$$= (\beta + (1/n)n(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e})$$

$$= (\beta + (\frac{1}{n}(\frac{1}{n})^{-1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e})) \quad (\text{trick})$$

$$\hat{\beta} = \beta + (\frac{1}{n}\mathbf{X}'\mathbf{X})^{-1})(\frac{1}{n}\mathbf{X}'\mathbf{e})$$

Proof of Consistency of OLS: Route 2, Step 2

$$\begin{aligned}\hat{\beta} &= \beta + \left(\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1}\left(\frac{1}{n}\mathbf{X}'\mathbf{e}\right) \\ \text{plim}\hat{\beta} - \beta &= \left(\text{plim}\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1}\text{plim}\frac{1}{n}\mathbf{X}'\mathbf{e} && \text{(Slutsky)} \\ &= \mathbf{Q}_{\mathbf{X}\mathbf{X}}^{-1}\mathbb{E}[\mathbf{X}\mathbf{e}] \\ &= \mathbf{Q}_{\mathbf{X}\mathbf{X}}^{-1}0\end{aligned}$$

The least squares estimator $\hat{\beta}$ is *consistent* for the projection coefficient β .

Key condition: Consistency requires $\mathbb{E}[\mathbf{x}\mathbf{e}] = 0$. When this fails (endogeneity), OLS is inconsistent — motivating instrumental variables (Lecture 13).

Asymptotic Normality: Algebraic Setup

Start from the sampling error decomposition:

$$\hat{\beta} - \beta = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i e_i \right) \quad (1)$$

Multiply both sides by $\sqrt{n}$:

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i e_i \right) \quad (2)$$

$$= \hat{Q}_{XX}^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i e_i \right) \quad (3)$$

Now we need the distribution of $\frac{1}{\sqrt{n}} \sum X_i e_i$. This is a sum of mean-zero iid random vectors — exactly the setting of the **multivariate CLT** from Lecture 10.

Asymptotic Normality: Applying CLT and Slutsky

$$\sqrt{n}(\hat{\beta} - \beta) = \underbrace{\hat{\mathbf{Q}}_{XX}^{-1}}_{\xrightarrow{P} \mathbf{Q}_{XX}^{-1}} \cdot \underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i e_i}_{\xrightarrow{d} N(0, \Omega)}$$

where $\Omega = \mathbb{E}[X_i e_i e_i' X_i'] = \mathbb{E}[X_i X_i' e_i^2]$.

By Slutsky's theorem (convergent in probability $\times$ convergent in distribution):

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} \mathbf{Q}_{XX}^{-1} \cdot N(0, \Omega) \tag{4}$$

$$= N(0, \mathbf{Q}_{XX}^{-1} \Omega \mathbf{Q}_{XX}^{-1}) \equiv \mathbf{V}_{\beta} \tag{5}$$

Same structure as Probit MLE (Lecture 9): $\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N(0, \mathcal{I}^{-1})$. In both cases, CLT gives the numerator; WLLN/CMT stabilize the denominator; Slutsky combines them.

Covariance of coefficient estimators under homoskedasticity

- Assume $Y = \beta_1 X_1 + \beta_2 X_2 + e$ with $E[ee'] = \sigma^2 I$.
- Suppose $E[X_1] = 0$, $E[X_2] = 0$, $\text{Var}[X_1] = 1$, $\text{Var}[X_2] = 1$, $\text{corr}[X_1, X_2] = \rho$

$$\mathbf{v}_{\beta}^0 = \sigma^2 \mathbf{Q}_{XX}^{-1} = \sigma^2 \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}^{-1} = \frac{\sigma^2}{1 - \rho^2} \begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix}$$

- If X_1 and X_2 are positively correlated, $\hat{\beta}_1$ and $\hat{\beta}_2$ are negatively correlated.

Heteroskedastic Covariance Matrix Estimation

- The asymptotic covariance matrix of $\sqrt{n}(\hat{\beta} - \beta)$ is $\mathbf{Q}_{XX}^{-1}\Omega\mathbf{Q}_{XX}^{-1}$.
- This is the **sandwich form** from Lecture 6 — now with its asymptotic justification.

Estimator:

$$\hat{\Omega} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \hat{e}_i^2$$

$$\hat{\mathbf{Q}}_{XX} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i'$$

$$\hat{\mathbf{V}}_{\beta}^{HC0} = \hat{\mathbf{Q}}_{XX}^{-1} \hat{\Omega} \hat{\mathbf{Q}}_{XX}^{-1}$$

This is the HC0 estimator. Recall from Lecture 6: HC1, HC2, HC3 apply finite-sample corrections. All are asymptotically equivalent.

Consistency of $\hat{\Omega}$

$$\begin{aligned}\hat{\Omega} &= \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \hat{e}_i^2 \\ &= \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \hat{e}_i^2 + \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' e_i^2 - \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' e_i^2 \\ &= \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' e_i^2 + \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' (\hat{e}_i^2 - e_i^2)\end{aligned}$$

$$\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' e_i^2 \xrightarrow{p} \mathbb{E}[\mathbf{x} \mathbf{x}' e^2] = \Omega \quad (\text{WLLN})$$

$$\hat{\Omega} \xrightarrow{p} \Omega$$

The Sandwich Pattern

The same “bread–meat–bread” structure appears throughout this course:

Model	Bread	Meat (Ω)	Simplification
OLS	Q_{XX}^{-1}	$\mathbb{E}[XX'e^2]$	$\sigma^2 Q_{XX}^{-1}$ if homosk.
Probit MLE	$\mathcal{I}^{-1}$	$\mathbb{E}[s_i s_i']$	$\mathcal{I}^{-1}$ if correct spec.
GMM	$(G'WG)^{-1}G'W$	$\mathbb{E}[g_i g_i']$	—

- Under “ideal” conditions (homoskedasticity / correct specification), the sandwich simplifies.
- Under misspecification, we need the full sandwich — **robust SEs**.
- Same `sandwich::vcovHC()` in R handles all cases.

Variances: Exact vs Asymptotic

Object	Formula	When to use
Exact variance (homosk.)	$(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{D}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}$ $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$	Fixed X , any n $+E[ee'] = \sigma^2 I$
Asy. variance (homosk.)	$\mathbf{Q}_{XX}^{-1}\Omega\mathbf{Q}_{XX}^{-1}$ $\sigma^2\mathbf{Q}_{XX}^{-1}$	Random X , n large $+E[e^2 X] = \sigma^2$

- The exact variance conditions on X ; the asymptotic variance averages over X .
- For large n : $\frac{1}{n}\mathbf{X}'\mathbf{X} \approx \mathbf{Q}_{XX}$, so they agree.
- In practice, we always estimate using the sandwich (or its homoskedastic special case).

From F Tests to Wald Tests

- We saw we can calculate an F test by forming a matrix of restrictions.
- The F distribution result depended on the homoskedasticity and normality of the errors.
- In large samples, we can instead rely on the asymptotic behavior of random variables and calculate a **Wald statistic**.
- Many test statistics are re-flavored Wald distances. Given a null that $E[\mathbf{q}] = \boldsymbol{\theta}$:

$$W = (\mathbf{q} - \boldsymbol{\theta})' [\widehat{\text{Var}}(\mathbf{q} - \boldsymbol{\theta})]^{-1} (\mathbf{q} - \boldsymbol{\theta})$$

- $W \sim \chi_J^2$ if $\mathbf{q}$ is Normal and $\text{Var}(\mathbf{q} - \boldsymbol{\theta}) = \boldsymbol{\Sigma}$
- $W \overset{a}{\sim} \chi_J^2$ with CLT on $\mathbf{q}$ and consistency for $\hat{\boldsymbol{\Sigma}}$

Mahalanobis Distance: Intuition

- Suppose $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. We want to measure how unusual an observation $\mathbf{x}$ is.
- In the scalar case:

$$D = \frac{x - \mu}{\sigma} \sim \mathcal{N}(0, 1) \quad \Rightarrow \quad D^2 \sim \chi_1^2$$

- In the vector case: transform the data to make it standard normal:

$$\mathbf{z} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{x} - \boldsymbol{\mu}) \sim \mathcal{N}(0, \mathbf{I})$$

- Then the squared norm of $\mathbf{z}$ gives:

$$\|\mathbf{z}\|^2 = (\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) = D^2$$

- This is the Mahalanobis distance: it tells us how far $\mathbf{x}$ is from $\boldsymbol{\mu}$ in standardized units, accounting for correlations and scale.

Wald Statistic as a Distance

- The Wald test generalizes the standardized t -statistic to higher dimensions:

$$\text{Univariate: } \frac{\hat{\beta} - \beta_0}{\text{SE}(\hat{\beta})} \sim \mathcal{N}(0, 1) \Rightarrow \left(\frac{\hat{\beta} - \beta_0}{\text{SE}(\hat{\beta})} \right)^2 \sim \chi_1^2$$

- In multivariate form:

$$W = (\mathbf{q} - \boldsymbol{\theta})' \hat{\boldsymbol{\Sigma}}^{-1} (\mathbf{q} - \boldsymbol{\theta})$$

- If $\mathbf{q}$ is consistent and asymptotically normal:

$$\sqrt{n}(\mathbf{q} - \boldsymbol{\theta}) \xrightarrow{d} \mathcal{N}(0, \boldsymbol{\Sigma}) \Rightarrow W \overset{a}{\sim} \chi_J^2$$

- Interpretation: Wald tests ask whether the observed $\mathbf{q}$ is “far” from $\boldsymbol{\theta}$, accounting for sampling variability.

Matrix form of Restrictions

- Suppose we have k parameters β and q restrictions with $q \leq (k)$. (Note $q = J$ in Greene, here k includes the intercept)
- Let $\mathbf{R}$ be a $q \times (k)$ matrix.
- We can express our null hypothesis as $H_0 : \mathbf{R}\beta = \mathbf{r}$.
- For example in the prior example where $\beta_2 = 0$ and $\beta_3 = \beta_4$

$$\mathbf{R} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}, \quad \mathbf{r} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Wald Test for Linear Restrictions

$$W = (R\hat{\beta} - r)'[R\hat{V}_{\hat{\beta}}R']^{-1}(R\hat{\beta} - r)$$
$$W \stackrel{a}{\sim} \chi_q^2$$

- If we use $\hat{V}_{\hat{\beta}} = s^2(\mathbf{X}'\mathbf{X})^{-1}$ (homoskedasticity + normality), then W/q is exactly the F statistic distributed $F_{q,n-k}$.
- Without normality, the CLT tells us W/q is approximately $F_{q,n-k}$.
- Without normality or homoskedasticity, use $\hat{V}_{\hat{\beta}}^{HC}$ (sandwich) and $W \stackrel{a}{\sim} \chi_q^2$.

The Wald test nests the F test as a special case. With robust SEs, it works without normality or homoskedasticity — only requiring consistency and asymptotic normality.

Wald Tests in R

```
library(sandwich); library(lmtest); library(car)

mod0 <- lm(prestige ~ income + education + women,
           data = Prestige)

# Wald test with robust SEs (chi-squared)
linearHypothesis(mod0,
  c("education = 0", "women = 0"),
  vcov = vcovHC, test = "Chisq")

# Equivalent: waldtest with nested models
mod1 <- lm(prestige ~ income, data = Prestige)
waldtest(mod1, mod0, vcov = vcovHC, test = "Chisq")
```

Both test $H_0 : \beta_{ed} = \beta_{women} = 0$ using the sandwich covariance.

Applying the Delta Method to OLS

Earlier we introduced the delta method in general: if $\hat{\theta}$ is asymptotically normal, then any smooth function $h(\hat{\theta})$ is also asymptotically normal.

Now we have the specific ingredients for OLS:

- $\hat{\beta}$ is asymptotically normal (proved in the previous sections)
- $\hat{\mathbf{V}}_{\hat{\beta}}$ is a consistent estimator of its covariance (sandwich or homoskedastic)
- For any differentiable $\theta = r(\beta)$, the gradient $\mathbf{R} = \left. \frac{\partial r}{\partial \beta} \right|_{\hat{\beta}}$ gives:

$$s(\hat{\theta}) = \sqrt{\mathbf{R}' \hat{\mathbf{V}}_{\hat{\beta}} \mathbf{R}}$$

This works for **any** differentiable function of $\hat{\beta}$ — linear or nonlinear.

Let's work through three examples of increasing complexity.

Running Example: Log Wage Regression

- $\log(\text{wage}) = \beta_1 \cdot \text{education} + \beta_2 \cdot \text{experience} + \beta_3 \cdot \text{experience}^2/100 + \beta_4 + e.$
- We can calculate:
 - Percentage return to education:

$$\theta_1 = 100\beta_1$$

- Percentage return to experience for those with 10 years of experience:

$$\theta_2 = 100\beta_2 + 20\beta_3$$

- Experience level which maximizes expected log wages:

$$\theta_3 = -50\beta_2/\beta_3$$

Each is a function $\theta = r(\beta)$. The first two are linear; the third is nonlinear. All three use the same delta method formula.

Example: Return to Education (Linear)

- Percentage return to education:

$$\theta_1 = 100\beta_1$$

- $R = \begin{pmatrix} 100 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

- Asymptotic standard error:

$$s(\hat{\theta}_1) = \sqrt{R' \hat{V}_{\hat{\beta}} R} = 100 \cdot \sqrt{\hat{\text{Var}}(\hat{\beta}_1)}$$

This is just rescaling — the SE scales linearly.

Example: Return to Experience at 10 Years (Linear)

- Percentage return to experience at 10 years:

$$\theta_2 = 100\beta_2 + 20\beta_3$$

- Gradient of θ_2 with respect to β :

$$\mathbf{R} = \begin{pmatrix} 0 \\ 100 \\ 20 \\ 0 \end{pmatrix}$$

- Asymptotic standard error:

$$s(\hat{\theta}_2) = \sqrt{\mathbf{R}' \hat{\mathbf{V}}_{\hat{\beta}} \mathbf{R}}$$

This is a linear combination of two coefficients — the SE depends on their covariance.

Example: Maximum Return to Experience (Nonlinear)

- Experience level which maximizes expected log wages:

$$\theta_3 = -50 \frac{\beta_2}{\beta_3}$$

- Gradient of θ_3 with respect to β :

$$\mathbf{R} = \begin{pmatrix} 0 \\ -50/\beta_3 \\ 50\beta_2/\beta_3^2 \\ 0 \end{pmatrix}$$

- Asymptotic standard error:

$$s(\hat{\theta}_3) = \sqrt{\mathbf{R}' \hat{\mathbf{V}}_{\hat{\beta}} \mathbf{R}}$$

This is nonlinear — $\mathbf{R}$ itself depends on β , estimated at $\hat{\beta}$. Same logic as the long-run elasticity from the gas consumption example.

The z-statistic

- The t-statistic is $T(\theta) = \frac{\hat{\theta} - \theta}{s(\hat{\theta})}$
- When dealing with asymptotic inference, this is called the z-statistic.

$$\begin{aligned} T(\theta) &= \frac{\hat{\theta} - \theta}{s(\hat{\theta})} \\ &= \frac{\sqrt{n}(\hat{\theta} - \theta)}{\sqrt{\hat{V}_\theta}} \\ &\rightarrow_d \frac{N(0, V_\theta)}{\sqrt{V_\theta}} \\ &= Z \sim N(0, 1) \end{aligned}$$

Under $H_0 : \theta = \theta_0$, reject at level α if $|T| > z_{\alpha/2}$. For $\alpha = 0.05$: $z_{0.025} = 1.96$.

Regression Intervals

- Linear model: $m(x) = x'\beta$
- Define target parameter: $\theta = r(\beta) = x'\beta$
- Estimate: $\hat{m}(x) = \hat{\theta} = x'\hat{\beta}$
- Gradient of $r(\beta)$: $R = \frac{\partial r}{\partial \beta} = x$
- Asymptotic standard error:

$$s(\hat{\theta}) = \sqrt{R'\hat{V}_{\hat{\beta}}R} = \sqrt{x'\hat{V}_{\hat{\beta}}x}$$

- 95% confidence interval for $\mathbb{E}[Y|X = x]$:

$$x'\hat{\beta} \pm 1.96 \cdot \sqrt{x'\hat{V}_{\hat{\beta}}x}$$

Example: Prestige $\sim$ Education

$$\beta_{ed} = 5.4, \beta_{cons} = -10.7, \hat{\mathbf{V}}_{\hat{\beta}} = \begin{bmatrix} 13.52 & -1.18 \\ -1.18 & .11 \end{bmatrix}$$

at 10 years of education:

$$\begin{aligned} & \mathbf{x}'\hat{\beta} \pm 1.96 \cdot \sqrt{\mathbf{x}'\hat{\mathbf{V}}_{\hat{\beta}}\mathbf{x}} \\ & (1 \ 10) \begin{pmatrix} -10.7 \\ 5.4 \end{pmatrix} \pm 1.96 \cdot \sqrt{(1 \ 10) \begin{bmatrix} 13.52 & -1.18 \\ -1.18 & .11 \end{bmatrix} \begin{pmatrix} 1 \\ 10 \end{pmatrix}} \\ & 43.3 \pm 1.96 \cdot \sqrt{0.87} \\ & 43.3 \pm 1.83 \end{aligned}$$

compare to 22 years of education

$$\begin{aligned} & 107 \pm 1.96 \cdot \sqrt{14.8} \\ & 107 \pm 7.5 \end{aligned}$$

R: Sandwich SEs and Delta Method

```
# Homoskedastic vs robust SEs
coeftest(mod0)                                # model-based
coeftest(mod0, vcov = vcovHC(mod0, type="HC1")) # sandwich

# Confidence interval for CEF at a point
x0 <- c(1, 5000, 12, 30) # intercept, income, ed, women
se_x0 <- sqrt(t(x0) %*% vcovHC(mod0) %*% x0)
pred <- sum(x0 * coef(mod0))
cat(pred, "+/-", 1.96 * se_x0)

# Delta method for nonlinear function (car package)
car::deltaMethod(mod0, "income / education")
```

Same tools as Lectures 6, 9, 10 — the sandwich and car packages handle the asymptotic theory automatically.

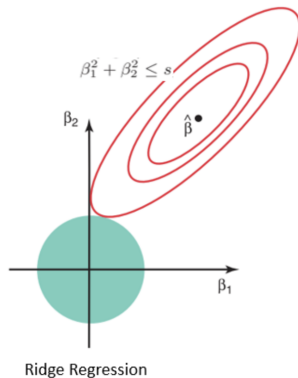
Application: Ridge Regression

- Suppose $Y = X'\beta + e$ with $\mathbb{E}[Xe] = 0$

$$\hat{\beta}_R = \left(\sum_{i=1}^n X_i X_i' + \lambda I_k \right)^{-1} \left(\sum_{i=1}^n X_i Y_i \right)$$

- Where $\lambda > 0$ is a constant, added to the diagonals of the covariance matrix.
- This is biased, shrinks $\hat{\beta}$.
- Arises from solving $\min_{\beta} (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) + \lambda(\beta'\beta - c)$

Application: Ridge Regression



Ridge is Consistent (via CMT)

$$\begin{aligned}\text{plim } \hat{\beta}_R &= \text{plim} \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' + \frac{1}{n} \lambda I_k \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i Y_i \right) \\&= \left(\text{plim} \frac{1}{n} \sum_{i=1}^n X_i X_i' + \text{plim} \frac{1}{n} \lambda I_k \right)^{-1} \left(\text{plim} \frac{1}{n} \sum_{i=1}^n X_i Y_i \right) \\&= (\mathbb{E}[X_i X_i'])^{-1} \mathbb{E}[X_i Y_i] \quad \text{since } \frac{\lambda}{n} \rightarrow 0 \text{ as } n \rightarrow \infty\end{aligned}$$

- As $n \rightarrow \infty$, the Ridge estimator converges in probability to the same plim as OLS.
- Biased in finite samples, consistent asymptotically — the canonical example of a

What We Proved Today

The OLS estimator $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is:

Property	Tool (from Lectures 10–11)	Result
Consistent	WLLN + CMT	$\text{plim } \hat{\beta} = \beta$
Asy. normal	Multivariate CLT + Slutsky	$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V_{\beta})$
Testable	Wald statistic	$W \overset{a}{\sim} \chi_q^2$
Robust	Sandwich estimator	$\hat{V}_{\beta}^{HC} \xrightarrow{p} V_{\beta}$

These results hold **without normality** — only requiring iid data with finite 4th moments and $\mathbb{E}[\mathbf{x}\mathbf{e}] = 0$.

Looking Ahead

What happens when $\mathbb{E}[\mathbf{x}e] \neq 0$?

- OLS is **inconsistent**: $\text{plim } \hat{\beta} = \beta + Q_{XX}^{-1} \mathbb{E}[Xe] \neq \beta$
- No amount of data can fix this — the bias does not vanish.

Lectures 13–14 (IV and 2SLS): Find instruments Z with $\mathbb{E}[Ze] = 0$ and $\mathbb{E}[ZX'] \neq 0$.

- Replace the OLS moment condition $\mathbb{E}[X(Y - X'\beta)] = 0$ with $\mathbb{E}[Z(Y - X'\beta)] = 0$
- Same asymptotic machinery: WLLN, CLT, Slutsky, sandwich — applied to a different estimating equation
- This is another instance of the GMM framework: $\mathbb{E}[g(W_i, \theta_0)] = 0$